

Homework Section 16.6

1. Determine whether the points $P(-2, 6, 5)$ and $Q(7, 3, 15)$ lie on the surface given by $\mathbf{r}(u, v) = \langle u + 3v, 1 + 4u - v, 5 + u + v \rangle$.
2. Find a Cartesian equation for the given parametric surface and identify it.
 - a) $\mathbf{r}(u, v) = (u - v)\mathbf{i} + (2 - v)\mathbf{j} + (4 + 2u - 4v)\mathbf{k}$
 - b) $\mathbf{r}(s, t) = s\mathbf{i} + t\mathbf{j} + (s^2 + t^2)\mathbf{k}$
 - c) $\mathbf{r}(u, v) = \langle u, u \sin 3v, u \cos 3v \rangle$
3. Find a parametric representation for the surface.
 - a) The hyperboloid $z = x^2 - 2y^2$.
 - b) The lower half of the ellipsoid $x^2 + 4y^2 + 9z^2 = 1$.
 - c) The sphere $x^2 + y^2 + z^2 = 9$.
4. Find an equation of the tangent plane to the given parametric surface at the specified point:
 $\mathbf{r}(u, v) = u \sin v \mathbf{i} + u^2 \mathbf{j} + 2u \cos v \mathbf{k}$; $u = 1, v = \pi$.
5. Find the area of the surface.
 - a) The part of the plane $2x + y + z = 6$ that lies in the first octant.
 - b) The part of the hyperbolic paraboloid $f(x, y) = x^2 - y^2$ that lies within the cylinder $x^2 + y^2 = 4$.
 - c) The part of the cone $z = \sqrt{x^2 + y^2}$ that lies between the planes $z = 1$ and $z = 4$.
 - d) The surface given by $\mathbf{r}(u, v) = uv\mathbf{i} + v^2\mathbf{j} + \frac{1}{2}v^2\mathbf{k}$, where $0 \leq u \leq 2, 0 \leq v \leq 1$.